EXPERIMENT-12

AIM:

To Implement echo clientserver using TCP/UDP sockets program.

CODE: (TCP)

```
import socket
import threading
#--- SERVER ---
def start_server():
    server = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    While True:
      data = conn.recv(1024).decode()
      if not data or data.lower() == 'quit':
          print("[SERVER] Connection closed by client.")
          break
      print(f"[SERVER] Received: {data}")
      conn.send(f"Server received: {data}".encode())
conn.close()
server.close()

#-- CLIENT- -
def start_client():
        client = socket.socket(socket.AF_INET,
                               socket.SOCK_STREAM)
        client.connect(('127.0.0.1', 5000))
        print("[CLIENT] Connected to server")
```

```
    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

#MAIN---

if __name__ == "__main__":
    #Run server in a background thread
    server_thread = threading.Thread(target=start_server,
                                     daemon=True)
    server_thread.start()

    #Run client in main thread
    start_client()
```

OUTPUT:

```
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to Server
[SERVER] Connected by ('127.0.0.1', 53458) Enter message (or
                                        'quit' to exit):
Hello
[SERVER] Received: Hello
[SERVER RESPONSE] Server received: Hello
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client
```

UDP:

```
import socket
import threading
#--SERVER--
def udp_server():
    S = socket.socket(socket.AF_INET,
                      socket.SOCK_DGRAM)
    S.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = S.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended
                  the connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        S.sendto(data, addr)
    S.close()
    print("[SERVER] Stopped.")

#---CLIENT---
def udp_client():
    C = socket.socket(socket.AF_INET,
                      socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message ('or 'quit' to
                    stop): ")
        C.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connection...")
            break
        data, _ = C.recvfrom(1024)
        print("Echo from server:", data.decode())
    C.close()
    print("[CLIENT] Stopped.")
```

```
#---MAIN---
if __name__ = "__main__":
    Threading.Thread (target = udp-server,
                      daemon = True).Start()

    udp_client()
```

OUTPUT:

Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000 Hi

[SERVER Received from ('127.0.0.1', 49870): Hi
Echo from server: Hi
Enter message (or 'quit' to exit): quit
[CLIENT] Stopped. [SERVER] Stopped.

RESULT:

Thus the program to implement echo client server using TCP/UDP socket was executed successfully.